Department of Computer Science and Engineering

Database Management Systems

Week 5 | DML and Join SQL Queries

Smart Hostel Management System -- A Case Study

Objective

To practise SQL DML (INSERT / UPDATE / DELETE), set operations (UNION / INTERSECT / EXCEPT), and JOIN queries (inner join, self join, left join, multi-table join, join with GROUP BY) on the Smart Hostel Management System database you designed in Lab 2 (ER Model) and implemented in Lab 3 (Relational Schema & DDL).

Outcome

By completing this lab, students will learn to:

- Perform INSERT, UPDATE, and DELETE operations using SQL DML.
- Use set operations such as UNION, INTERSECT, and EXCEPT or their supported equivalents.
- Perform inner, self, left, and multi-table joins.
- Use GROUP BY and aggregate functions with joins.
- Write SQL queries based on the relational schema of the Smart Hostel Management System.
- Execute and verify SQL queries using MySQL.

Before You Begin

- Create (or re-create) all 11 tables exactly as specified below. Use your own appropriate SQL syntax and data types -- the specification tells you what each table and column must be. Populate your tables with data as per depicted below. Do not redesign the case study or change the schema from what is specified.
- Keep this database as-is after finishing the lab -- see Deliverables below.

- Execute every SQL query in MySQL and verify the result before recording the output. Include the SQL query and a screenshot showing the executed query and its output for each question.
- Write your queries in notepad after executing in mysql, you can save the notepad file as .sql for submission.

Deliverables

- A single PDF containing, in order: your schema creation screenshots, your inserted data screenshots(using select), and then each of the 20 questions with the SQL query you wrote and a screenshot of its output and the SQL file with the complete code which you implemented .
- File names:
  - <SRN>_Lab04_DML_Join.pdf
  - <SRN>_Lab04_DML_Join.sql
- Keep the database as-is after submission -- the same schema and the data you build in this lab will be reused in upcoming labs on nested/correlated queries, procedures and functions, and triggers/transactions.
- Before running any queries, execute this in your terminal :
  - PROMPT (SRN@\u) [\d]> (replace SRN with your actual SRN)

```
mysql> PROMPT (PES1UG23CS624@\u) [\d]>
PROMPT set to '(PES1UG23CS624@\u) [\d]>'
(PES1UG23CS624@root) [(none)]>create database hostel;
Query OK, 1 row affected (1.14 sec)

(PES1UG23CS624@root) [(none)]>use hostel;
Database changed
(PES1UG23CS624@root) [hostel]>
```

Tasks to be executed:

- **Task 1: Database Preparation**
  - Create/re-create the 11 tables.
  - Populate sample data.

- Verify the tables/data.

- **Task 2: DML and Set Operations: PART A**
  - Questions 1–10

- **Task 3: Join Queries: PART B**
  - Questions 1–10

Task 1: Required Schema

Implement the following 11 tables. Primary keys, foreign keys and constraints must match what is described -- pick sensible data types and constraint syntax for whichever SQL platform you are using.

Hostel

- Hostel_ID — primary key
- Hostel_Name — mandatory
- Location — mandatory
- Capacity — mandatory, default 0, must be non-negative

Staff

- Staff_ID — primary key
- Staff_Name — mandatory
- Salary — mandatory, must be greater than 0
- Designation — mandatory, default 'Support Staff'
- Hostel_ID — mandatory, foreign key referencing HOSTEL
- Supervisor_ID — optional, foreign key referencing STAFF(Staff_ID)
- Deleting a supervisor should set Supervisor_ID of their subordinates to NULL.

Room

- Hostel_ID — part of the composite primary key and foreign key referencing HOSTEL

- Room_Number — part of the composite primary key
- Floor — mandatory, default 0
- A room is uniquely identified within its hostel.
- Deleting a hostel should delete its rooms.

Bed

- Hostel_ID, Room_Number, and Bed_Number — composite primary key
- Status — mandatory, default 'Vacant'
- Status must be either 'Occupied' or 'Vacant'
- (Hostel_ID, Room_Number) — composite foreign key referencing ROOM
- Deleting a room should delete its associated beds.

Student

- Student_ID — primary key
- First_Name — mandatory
- Middle_Name — optional
- Last_Name — mandatory
- Gender — mandatory
- DOB — mandatory
- Email — mandatory and unique
- Age should not be stored as a separate column; it should be derived from DOB when required.

Student_Phone

- Student_ID and Phone_Number — composite primary key
- Student_ID — foreign key referencing STUDENT
- Deleting a student should delete their phone numbers.

Facility

- Facility_ID — primary key
- Facility_Name — mandatory
- Operating_Hours — default '24x7'

- Hostel_ID — mandatory, foreign key referencing HOSTEL
- Deleting a hostel should delete its facilities.

Uses

- Student_ID, Facility_ID, and Usage_Date — composite primary key
- Student_ID — foreign key referencing STUDENT
- Facility_ID — foreign key referencing FACILITY
- Deleting a student or facility should delete the corresponding usage records.

Allocate

- Allocation_ID — primary key
- Student_ID — mandatory, foreign key referencing STUDENT
- Hostel_ID and Room_Number — mandatory composite foreign key referencing ROOM
- Allocation_Date — mandatory
- Check_in_Date — mandatory
- Check_out_Date — optional; NULL indicates that the student is currently residing in the room
- Deleting a student or room should delete the corresponding allocation records.

Payment

- Payment_ID — primary key
- Student_ID — mandatory, foreign key referencing STUDENT
- Payment_Date — mandatory
- Amount — mandatory, must be greater than 0
- Payment_Type — mandatory

Receipt

- Receipt_ID — primary key
- Payment_ID — mandatory and unique, foreign key referencing PAYMENT
- Receipt_Date — mandatory
- The unique Payment_ID ensures that each payment can have at most one receipt.

Sample Data -- Guidelines

Populate Tables with Sample Data

Populate the tables using the following sample data.

HOSTEL

Hostel_ID	Hostel_Name	Location	Capacity
1	Sunrise Hostel	North Campus	120
2	Blue Ridge Hostel	South Campus	90
3	Lake View Hostel	East Campus	60

STAFF

Staff_ID	Staff_Name	Designation	Salary	Hostel_ID	Supervisor_ID
1	Ramesh Iyer	Warden	45000	1	NULL
2	Sunil Kumar	Security	22000	1	1
3	Geeta Rao	Cleaner	18000	1	1
4	Anita Sharma	Warden	46000	2	NULL
5	Mohan Das	Security	21000	2	4
6	Priya Menon	Maintenance	25000	2	4
7	Kiran Bose	Warden	44000	3	NULL

8	Farhan Sheikh	Cleaner	17500	3	7

ROOM

Hostel_ID	Room_Number	Floor
1	101	1
1	102	1
1	201	2
2	101	1
2	102	1
3	101	1

BED

Hostel_ID	Room_Number	Bed_Number	Status
1	101	1	Occupied
1	101	2	Occupied
1	102	1	Occupied
1	102	2	Vacant
1	201	1	Occupied
1	201	2	Vacant
2	101	1	Occupied
2	101	2	Vacant
2	102	1	Occupied

2	102	2	Occupied
3	101	1	Occupied
3	101	2	Vacant

STUDENT

Student_ID	First_Name	Middle_Name	Last_Name	Gender	DOB	Email
1	Aditya	NULL	Verma	M	2005-03-14	aditya.verma@pes.edu
2	Neha	R	Kulkarni	F	2005-07-22	neha.kulkarni@pes.edu
3	Rohan	NULL	Shetty	M	2004-11-05	rohan.shetty@pes.edu
4	Sanya	K	Malhotra	F	2005-01-30	sanya.malhotra@pes.edu
5	Vikram	NULL	Nair	M	2004-09-18	vikram.nair@pes.edu
6	Ishita	P	Bansal	F	2005-05-09	ishita.bansal@pes.edu
7	Arjun	NULL	Reddy	M	2005-02-27	arjun.reddy@pes.edu
8	Meera	S	Pillai	F	2004-12-12	meera.pillai@pes.edu

- **STUDENT_PHONE**

Student_ID	Phone_Number
1	9876500001
1	9876500011

2	9876500002
3	9876500003
4	9876500004
4	9876500014
5	9876500005
6	9876500006
7	9876500007
8	9876500008

- **FACILITY**

Facility_ID	Facility_Name	Operating_Hours	Hostel_ID
1	Wifi Lounge	24 Hours	1
2	Laundry Room	06:00-20:00	1
3	Gym	05:00-22:00	2
4	Study Room	24 Hours	2
5	Common Room	08:00-23:00	3

USES

Student_ID	Facility_ID	Usage_Date
1	1	2026-01-20
2	1	2026-01-20

3	1	2026-01-22
1	2	2026-01-21
4	3	2026-01-20
5	3	2026-01-21
4	4	2026-01-25
6	4	2026-01-22
7	5	2026-01-23

ALLOCATION

Allocation_ID	Student_ID	Room_Number	Allocation_Date	Check_in_Date	Check_out_Date
1	1	101	2026-01-05	2026-01-06	NULL
2	2	101	2026-01-05	2026-01-06	NULL
3	3	102	2026-01-05	2026-01-06	NULL
4	4	101	2026-01-05	2026-01-07	NULL
5	5	102	2026-01-06	2026-01-07	NULL
6	6	101	2026-01-06	2026-01-07	NULL
7	7	101	2026-01-07	2026-01-08	NULL
8	3	201	2025-09-01	2025-09-02	2025-12-20

PAYMENT

Payment_ID	Student_ID	Payment_Date	Amount	Payment_Type
1	1	2026-01-10	45000	Online
2	2	2026-01-11	45000	Cash
3	3	2026-01-12	42000	Card
4	4	2026-01-13	48000	Online
5	5	2026-01-13	48000	Cheque
6	6	2026-01-14	48000	Online
7	7	2026-01-15	44000	Cash
8	1	2026-01-15	45000	Online

RECEIPT

Receipt_ID	Payment_ID	Receipt_Date
1	1	2026-01-10
2	2	2026-01-11
3	3	2026-01-12
4	4	2026-01-10
5	5	2026-01-13
6	6	2026-01-14
7	7	2026-01-15

Task 2: Part A -- DML and Set Operations (10 Questions)

Q1. Insert a new student, Kabir S. Oberoi (Male, DOB 2005-08-19, email kabir.oberoi@pes.edu), into Student.

Q2. In a single INSERT statement, add both of Kabir's phone numbers -- 9876500009 and 9876500019 -- to Student_Phone.

Q3. In a single INSERT statement, record three new facility usage events: Kabir uses the WiFi Lounge (Facility_ID 1) on 2026-08-22, Kabir uses the Common Room (Facility_ID 5) on 2026-08-23, and Vikram Nair (Student_ID 5) uses the Gym (Facility_ID 3) on 2026-08-22.

Q4. The university approves a 10% salary hike for every Warden across all hostels. Update Staff accordingly.

Q5. Rohan Shetty (Student_ID 3) checks out of Room 102, Hostel 1 on 2026-08-20. Close his current allocation record and free up his bed (Bed_Number 1 in that room).

Q6. Record a new payment of 45000 (Online) made by Kabir on 2026-08-22, and the receipt generated for it on the same date.

Q7. Delete every payment that does not yet have a matching receipt. Do this with a subquery -- do not hard-code any Payment_ID.

Q8. List the names of all staff members working in Hostel 1 along with the full names of all students currently allocated to Hostel 1. Combine both result sets using UNION ALL. Display the name and role (Staff or Student) for each record.

Q9. Using INTERSECT, find the Student_IDs of students who have both made at least one payment and used at least one facility.

Q10. Using EXCEPT, find every student who has never made a payment.

Task 2: Part B -- Join Queries (10 Questions)

Q1. List the Student_ID, full name, hostel name and room number of every student who currently has an active allocation (Check_out_Date IS NULL).

Q2. List every staff member along with the name of their supervisor (show NULL/blank for staff with no supervisor).

Q3. List every payment together with the paying student's name and the corresponding receipt number, if one has been generated.

Q4. Find every student who has never used any hostel facility.

Q5. For each hostel, count how many students currently reside in it (include hostels with zero current residents).

Q6. List every facility usage record as: facility name, student name, usage date -- ordered by facility, then date.

Q7. For every room, report how many beds in it are currently vacant.

Q8. List every staff member with the name and location of the hostel they work at.

Q9. For each hostel, compute the total fee revenue collected from its currently-residing students.

Q10. List every facility together with how many times it has been used, including facilities that have never been used.

Q11. Write a query using a RIGHT JOIN between HOSTEL and FACILITY to list all facilities along with the name of the hostel providing each facility.

Q12. Write a query using a LEFT JOIN to list all students (`First_Name`, `Last_Name`, `Email_Address`) along with their assigned `Room_Number`. Ensure that students who do not have an allocated room, such as Rohan, are also included in the result.